

ENTERPRISE SECURITY ASSESSMENT LAB

AWS FREE TIER / BILLING VERIFICATION

Professional methodology report for post-lab billing and cost-control validation

Module	Cloud Security Assessment
Document Type	Individual Methodology Report
Phase	Phase 8
Evidence Source	14-aws-free-tier-cleanup-check.png
Publication Type	GitHub Portfolio - Redacted Lab Version
Prepared For	04-cloud-security/README.md documentation set
File Name	08-aws-free-tier-billing-verification-methodology-report.pdf

1. Executive Purpose

Technical exploitation alone is not enough for cloud security work. A mature cloud lab must include billing awareness and cleanup verification. This report documents the post-lab billing review that confirms no unexpected active resources or charges were ignored.

2. Scope and Evidence Mapping

Testing objective	Confirm that AWS Free Tier / billing dashboard was checked after CloudGoat cleanup.
Cloud platform	Amazon Web Services (AWS)
Lab context	CloudGoat / authorized vulnerable cloud lab environment
Primary tools	AWS Console, Billing and Cost Management, Free Tier dashboard
Evidence file(s)	14-aws-free-tier-cleanup-check.png
Result type	Cost and cleanup verification completed

3. Embedded Evidence Screenshot(s)

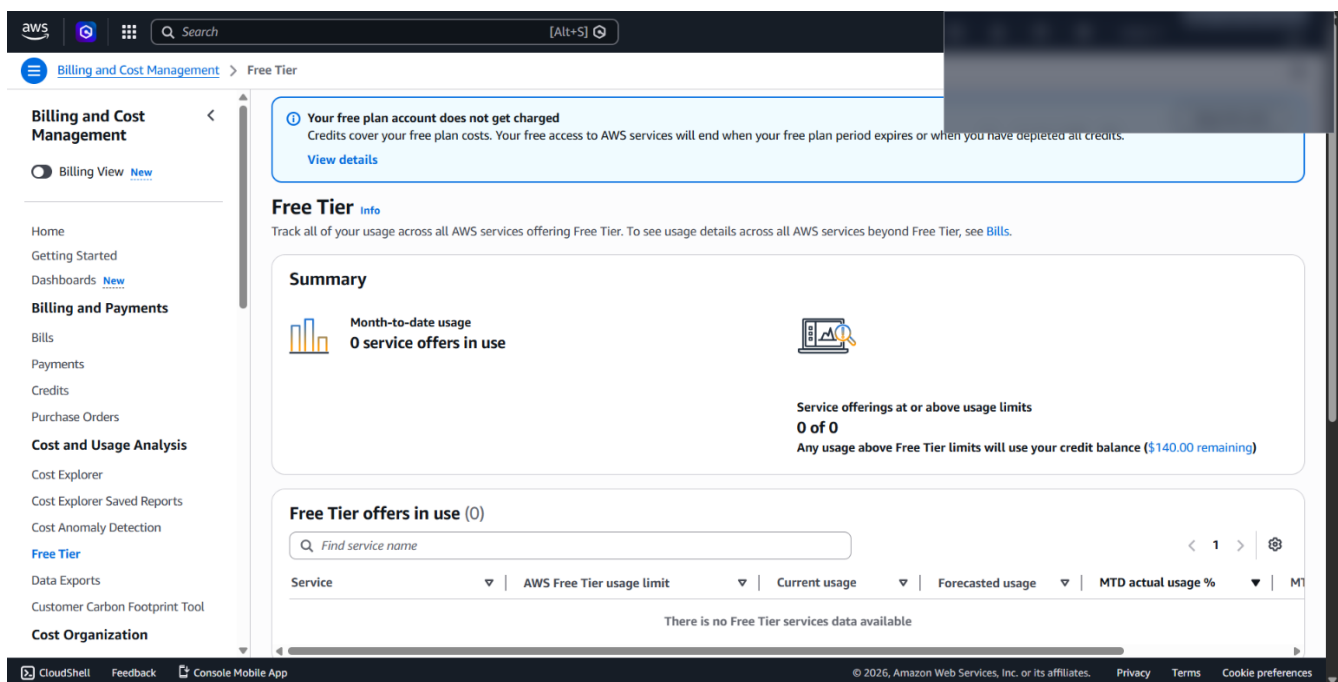


Figure 1 - AWS Free Tier / billing dashboard review after cleanup; account details redacted where applicable.

4. Methodology and Execution Workflow

The tester opens AWS Billing and Cost Management, reviews Free Tier or Bills, and checks whether the lab created unexpected ongoing cost. This is especially important after EC2, S3, IAM, and CloudGoat activity.

Command / workflow used

```
AWS Console
→ Billing and Cost Management
→ Free Tier / Bills
→ Confirm no unexpected active resources or charges
```

Execution notes

- Use budget alerts before running cloud labs.
- Check billing after every lab session, not only at month end.
- Review Cost Explorer or Bills if Free Tier usage looks unusual.
- Keep evidence redacted because billing pages may reveal account or payment context.

5. Sanitized Tool Output Style

The following output example is sanitized. It documents the expected evidence style without exposing AWS account IDs, ARN values, AccessKeyId values, SecretAccessKey values, session tokens, private bucket names, or billing details.

```
Billing verification completed.
No unexpected active lab costs documented in the evidence screenshot.
Account and private billing details redacted for GitHub publication.
```

6. Professional Interpretation

The evidence demonstrates that cloud lab work was handled responsibly. Recruiters can see that the project includes not only attack-path validation but also cost control and cleanup verification.

7. Risk Analysis

Risk Area	Explanation
Unexpected cost	Cloud labs can create charges through EC2, storage, data transfer, and managed services.
Operational maturity	Billing verification shows responsible use of real cloud infrastructure.
Account privacy	Billing screenshots can expose private account details and should be redacted.

8. Defensive Recommendations

- Create budget alerts before testing.
- Use a dedicated lab account.
- Destroy scenarios immediately after use.
- Review billing dashboard after every session.
- Document cleanup and cost verification in the project README.

9. GitHub Redaction and Publication Guidance

Before publishing this document or the underlying screenshot evidence, verify that the following values are removed, blurred, or replaced with placeholders. This is especially important because cloud evidence can contain values that are valid beyond the lab session if not rotated or destroyed.

Sensitive value	Required handling
AWS Account ID	Blur or replace with <REDACTED_ACCOUNT_ID>.
ARN values	Redact account number and private role/user names where needed.
AccessKeyId / SecretAccessKey	Must never be visible. Rotate immediately if exposed.

Sensitive value	Required handling
SessionToken	Must be fully redacted because temporary credentials can still be abused during validity period.
Bucket names / resource names	Redact if they identify private infrastructure or are still active.
Billing details	Hide account details, payment information, invoices, and private cost data.

10. Evidence Quality Checklist

- The screenshot is readable and aligned within the page margins.
- The evidence filename matches the GitHub link exactly.
- The command/output supports the stated result type.
- No AWS credentials, session tokens, account numbers, or private resource identifiers are visible.
- The report explains both the offensive testing value and the defensive remediation value.